ESAME 16-06-25

TRACCIA B

① Si studi la convergenza e l'assoluta convergenza della

$$\sum_{n=0}^{+\infty} (-1)^n \frac{n+4}{n^2+2}$$

È una serie a segno alterno

Studiamo l'assoluta convergenza

$$\sum_{n=0}^{+\infty} \frac{n+4}{n^2+2}$$

SERIE A TERMINI POSITIVI

Usiamo il criterio del confronto asintotico

$$\lim_{n\to+\infty} \frac{n+4}{n^2+2} = \lim_{n\to+\infty} \frac{n}{n^2} = \lim_{n\to+\infty} \frac{1}{n}$$

La serie

$$\sum_{n=0}^{+\infty} \frac{n+4}{n^2+2}$$ ha lo stesso carattere di

$$\sum_{n=1}^{\infty} \frac{1}{n}$$ che diverge

Allora $$\sum_{n=0}^{+\infty} (-1)^n \frac{n+4}{n^2+2}$$ non converge assolutamente

Studio

$$\sum_{n=0}^{+\infty} (-1)^n \frac{n+4}{n^2+2} \qquad a_n = \frac{n+4}{n^2+2}$$

tramite il criterio di Liebnitz

Per $\{a_n\}_{n\in\mathbb{N}}$ devono valere le ipotesi del

criterio di LEIBNIZ

(1) $a_n \geq 0 \quad \forall n \in \mathbb{N}$ ✓

(2) $\lim_{n\to+\infty} a_n = \lim_{n\to+\infty} \frac{n+4}{n^2+2} = 0$ ✓

$\sim \frac{1}{n}$

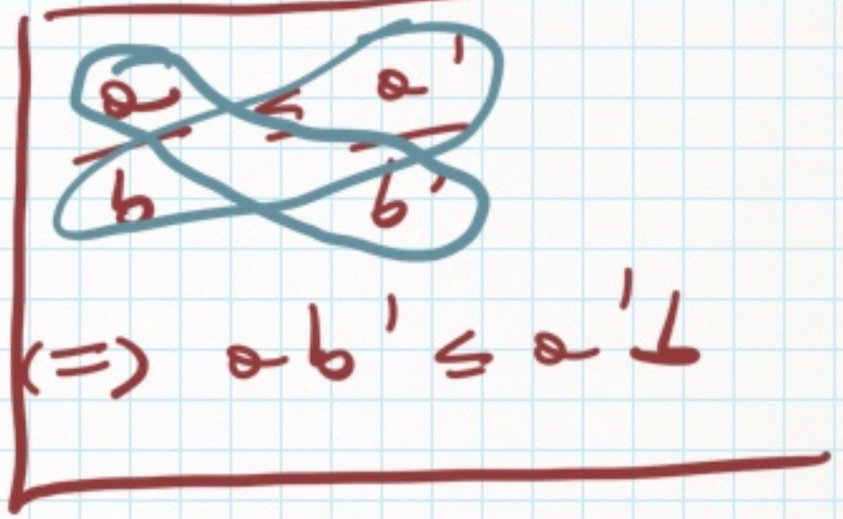
(3) $\exists\, n_0 \in \mathbb{N}$ t.c. $\forall n \geq n_0$

$$a_{n+1} \leq a_n \,, \text{ o sia}$$

$$\frac{n+1+4}{(n+1)^2+2} \leq \frac{n+4}{n^2+2}$$

ovvero

$$\frac{n+5}{n^2+2n+3} \leq \frac{n+4}{n^2+2}$$



$\Leftrightarrow a b' \leq a' b$

o sia

$$(n+5)(n^2+2) \leq (n+4)(n^2+2n+3)$$

DA DIMOSTRARE

$$(n+5)(n^2+2) \leq (n+4)(n^2+2n+3)$$

$$\Leftrightarrow \cancel{n^3} + 2n + 5n^2 + 10 \leq \cancel{n^3} + 2n^2 + 3n + 4n^2 + 5n + 12$$

$$\Leftrightarrow 0 \leq n^2 + 3n + 2 \Leftrightarrow n^2 + 3n + 2 \geq 0$$

$n \geq 0 \Leftrightarrow a_{n+1} \leq a_n \quad \forall n \geq 0$

La serie

$$\sum_{n=0}^{+\infty} (-1)^n \frac{n+4}{n^2+2}$$ converge mais

non converge absolument

② Si calcola

$$\int \frac{3x+2}{x^2-9}\,dx$$

FRATTI SEMPLICI

$x^2-9=0 \iff x=3 \vee x=-3$

$$\boxed{x^2-9=(x+3)(x-3)}$$

$$\left|\frac{3x+2}{x^2-9}\right| = \frac{A}{x+3}+\frac{B}{x-3} = \frac{Ax-3A+Bx+3B}{x^2-9} =$$

$$= \left|\frac{(A+B)x \;\; -3A+3B}{x^2-9}\right|$$

$$\begin{cases} A+B=3 \\ -3A+3B=2 \end{cases}$$

$$\begin{cases} A=3-B \\ -3(3-B)+3B=2 \end{cases}$$

$$\begin{cases} A=3-B \\ -9+3B+3B=2 \end{cases}$$

$$\begin{cases} A=3-\frac{11}{6}=\frac{7}{6} \\ B=\frac{11}{6} \end{cases}$$

Allora

$$\frac{3x+2}{x^2-9}=\frac{7}{6}\cdot\frac{1}{x+3}+\frac{11}{6}\cdot\frac{1}{x-3}$$

$$\int\frac{3x+2}{x^2-9}\,dx=\frac{7}{6}\int\frac{1}{x+3}\,dx+\frac{11}{6}\int\frac{1}{x-3}\,dx$$

$$=\frac{7}{6}\ln|x+3|+\frac{11}{6}\ln|x-3|+c$$

3.

$$\lim_{n\to+\infty} \frac{(n+3)!}{3^n + n! + \sqrt{n^4+2}}$$

$n!$ è un infinito di ordine superiore a 3^n e $\sqrt{n^4+2}$ e quindi

$$3^n + n! + \sqrt{n^4+2} \sim n!$$

Allora

$$\lim_{n\to+\infty} \frac{(n+3)!}{3^n + n! + \sqrt{n^4+2}} = \lim_{n\to+\infty} \frac{(n+3)!}{n!} =$$

$$= \lim_{n\to+\infty} \frac{\cancel{n!}\,(n+1)(n+\nu)(n+3)}{\cancel{n!}} =$$

$$= \lim_{n\to+\infty} n^3 = +\infty$$

$$\lim_{n\to+\infty} \underbrace{\left(1-\cos\left(\tfrac{2}{n}\right)\right)}_{\sim \frac{2}{n^2}} \underbrace{\left(\sqrt{2n^4+4n+1}\right)}_{\sim \sqrt{2}\cdot n^2}$$

$$= [0 \cdot +\infty]$$

$$\sqrt{2n^4+4n+1} \sim \sqrt{2n^4} = \sqrt{2}\cdot n^2$$

$$1-\cos x \sim \frac{x^2}{2} \qquad x \to 0$$

$$1-\cos\left(\frac{2}{n}\right) \sim \frac{1}{2}\cdot\left(\frac{2}{n}\right)^2 \qquad \frac{2}{n} \to 0$$

$$= \frac{1}{2}\,\frac{4}{n^2} = \frac{2}{n^2}$$

Allora

$$\lim_{n\to+\infty}\left(1-\cos\left(\frac{2}{n}\right)\right)\sqrt{2n^4+4n^3+1} =$$

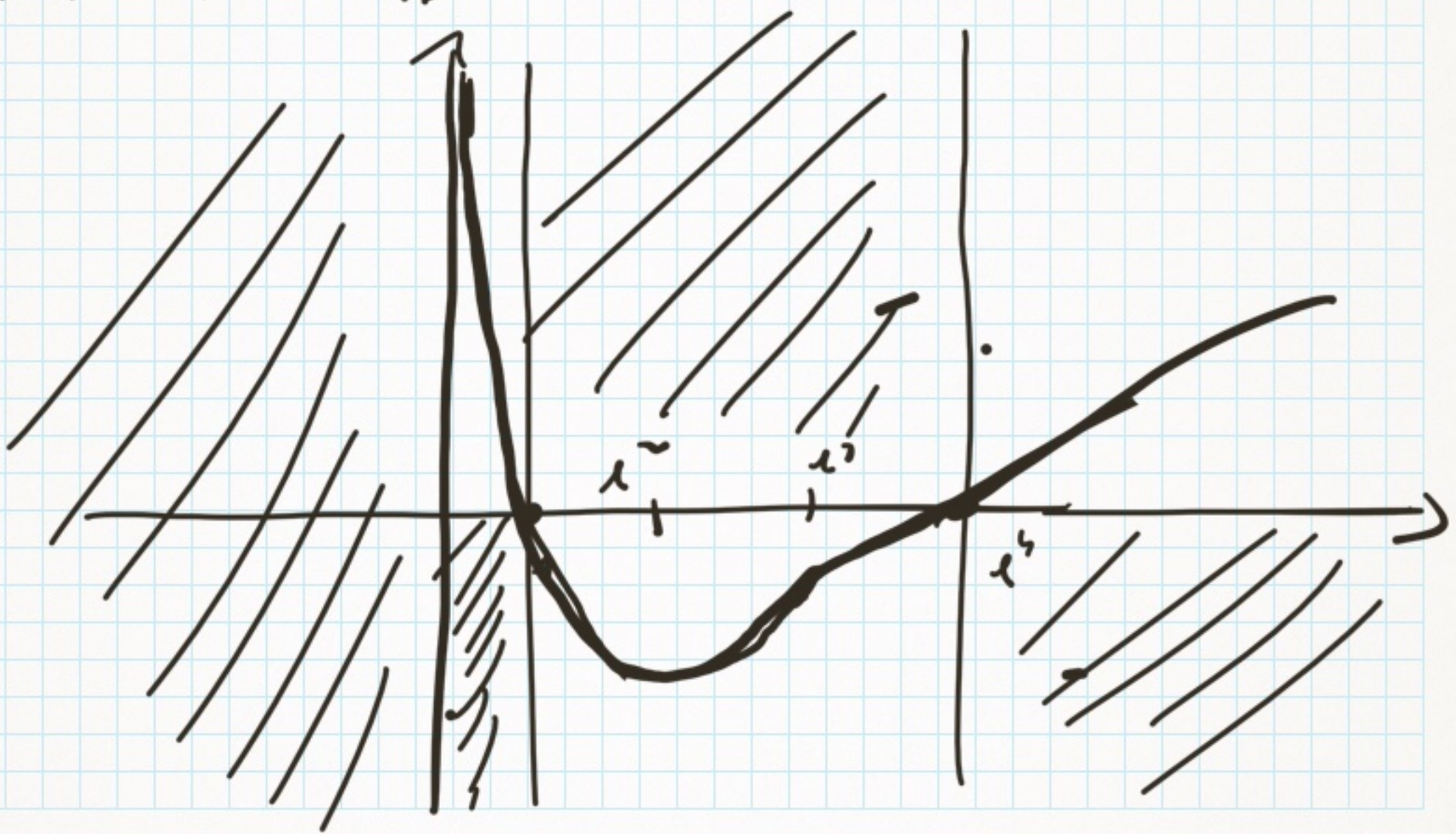
$$= \lim_{n\to+\infty} \frac{2}{n^2}\cdot\sqrt{2}\,n^2 = 2\sqrt{2}$$

Studiare la funzione

$$f(x) = \ln^2 x - 4 \ln x$$

$$\begin{array}{c} \\ 0 \\ \hline \leftarrow \quad \rightarrow \end{array}$$

DOMINIO: $x > 0$



DOMINIO NON SIMMETRICO $\Rightarrow$ f

NON È NÉ PARI NÉ DISPARI

NON CI SONO INTERSEZIONI CON $x=0$

POSITIVITÀ E INTERSEZIONI ASSE x

$f(x) \geq 0$

$\ln^2 x - 4 \ln x \geq 0$

$t = \ln x$

$t^2 - 4t \geq 0 \iff t \leq 0 \vee t \geq 4$

$\ln x \leq 0 \quad \vee \quad \ln x \geq 4$

$$\ln x \leq 0 \iff \begin{cases} x > 0 \\ x \leq e^0 = 1 \end{cases}$$

$$0 < x \leq 1$$

$$f(1) = 0$$

$$\ln x \geq 4 \implies x \geq e^4$$

$$f(e^4) = 0$$

$$f(x) \geq 0 \iff 0 < x \leq 1 \ \vee \ x \geq e^4$$

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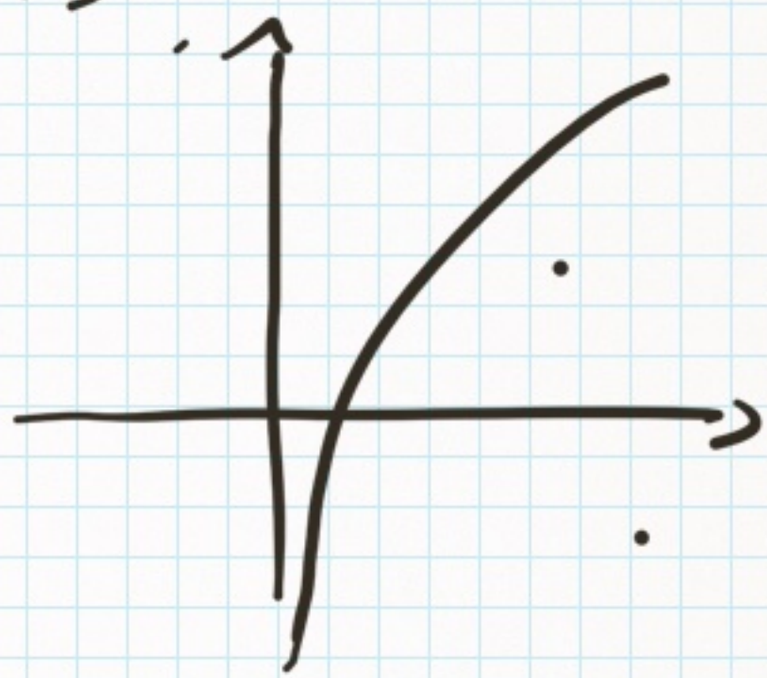
ASINTOTI

$$\lim_{x\to 0^+} \ln^2 x - 4\ln x =$$

$$= (-\infty)^2 - 4(-\infty)$$

$$= +\infty + \infty = +\infty$$

$$\lim_{x\to 0^+} \ln x = -\infty$$



$x = 0$ è asintoto verticale sx

$$\lim_{x\to+\infty} \ln^2 x - 4\ln x = \lim_{x\to+\infty} \ln^2 x = +\infty$$

non c'è asintoto orizzontale

asintoto obliquo

$$\lim_{x\to+\infty} \frac{f(x)}{x} = \lim_{x\to+\infty} \frac{\ln^2 x(-4\ln x)}{x} =$$

$$= \lim_{x\to+\infty} \frac{\ln^2 x}{x} = 0$$

non c'è asintoto obliquo

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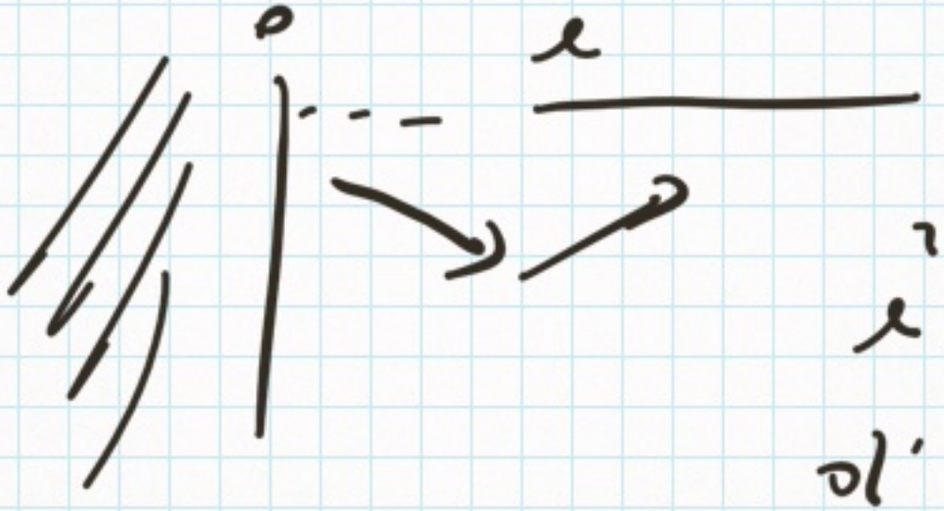
$$f(x) = \ln^2 x - 4\ln x$$

$$f'(x) = 2 \cdot \ln x \cdot \frac{1}{x} - \frac{4}{x} =$$

$$= \left(\frac{2}{x}\right)(\ln x - 2) \qquad \boxed{x > 0}$$

$$f'(x) \geq 0 \iff \ln x - 2 \geq 0$$

$$x \geq e^2$$



$x = e^2$ è punto di min

$$f'(x) = 2\,\frac{(\ln x - 2)}{x}$$

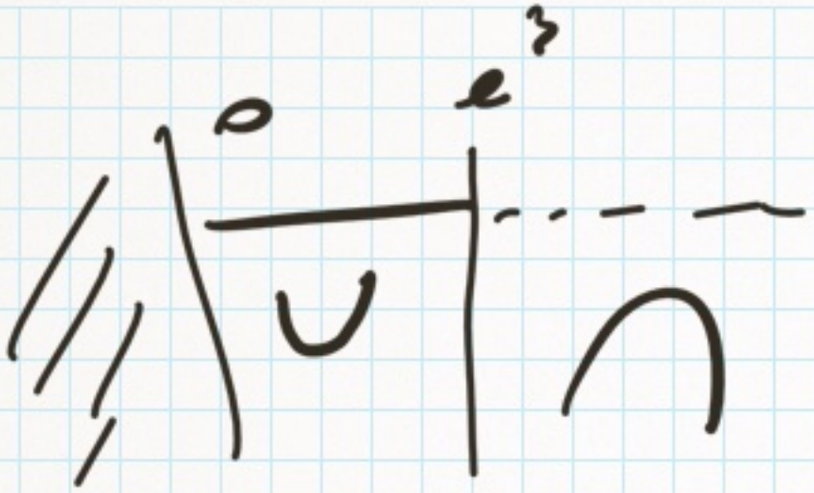
$$f''(x) = 2\,\frac{\frac{1}{x}\cdot x - (\ln x - 2)\cdot 1}{x^2} =$$

$$= 2\,\frac{1 - \ln x + 2}{x^2} = -\frac{\ln x + 3}{\underbrace{x^2}_{>0}} \geq 0$$

$$\Leftrightarrow \ominus \ln x + 3 \geq 0 \Leftrightarrow \ln x - 3 \leq 0 \,.$$

$$\Leftrightarrow \ln x \leq 3 \overset{0<}{\Leftrightarrow} x \leq e^3$$

$$f''(x) \geq 0 \Leftrightarrow 0 < x \leq e^3$$



$x = e^3$ è un punto di flesso

$$\sum_{n=1}^{\infty} \ln\left(1 + \frac{3}{n^5}\right) \cdot (n^3 + n) \sim$$

$$\sum_{n=1}^{\infty} \frac{3}{n^5} \cdot n^3 = \sum_{n=1}^{\infty} \frac{3}{n^2} \quad \text{converge}$$

.

$$\sum_{n=0}^{+\infty} \frac{n^2 \cdot 2^n}{3^n + n}$$

RAPPO PSO O RADICE

$$\frac{n^2 \cdot 2^n}{3^n + n} \sim \frac{n^2 \cdot 2^n}{3^n}$$

RAPPO PSO

O RADICE